Climate policies enhance the environmental sustainability of green chemicals

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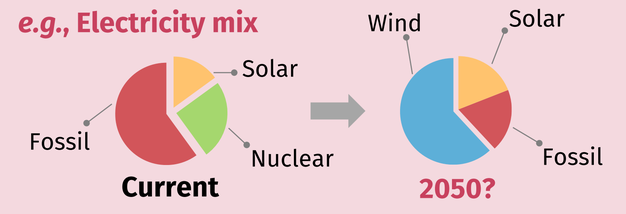
Introduction

- The chemical industry releases 5.6 Gt CO_2-eq yr^{-1} (10% of global emissions) due to its reliance on fossil fuels.[1]
- Switching to low-carbon hydrogen derived from solar or wind-powered water electrolysis can substantially reduce emissions for ammonia production.
- Current environmental studies do not consider the future evolution of power, materials, and transportation sectors under different climate policy scenarios.

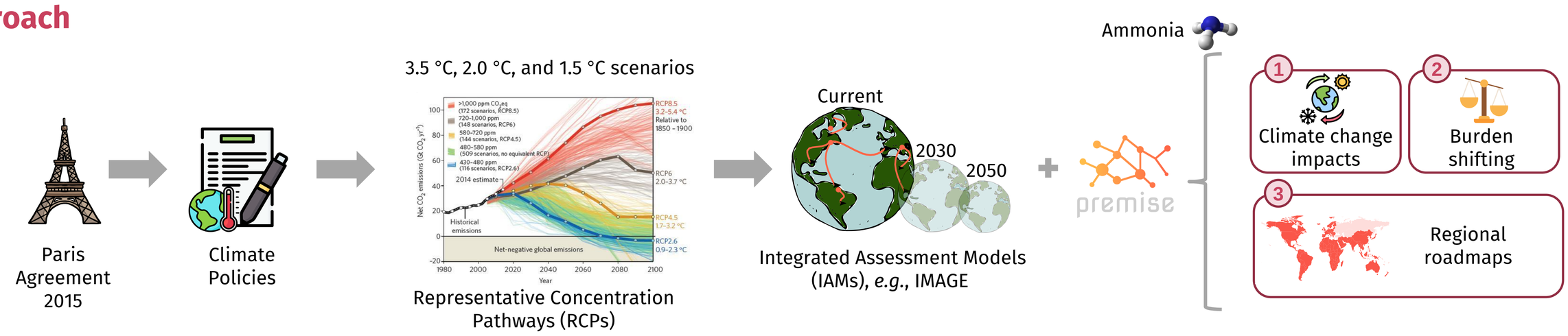
Goal

- Demonstrate how climate policies, affecting all components of chemical supply chain can synergistically drive sustainability in the industry.
- **How do the different production routes compete in the future?**

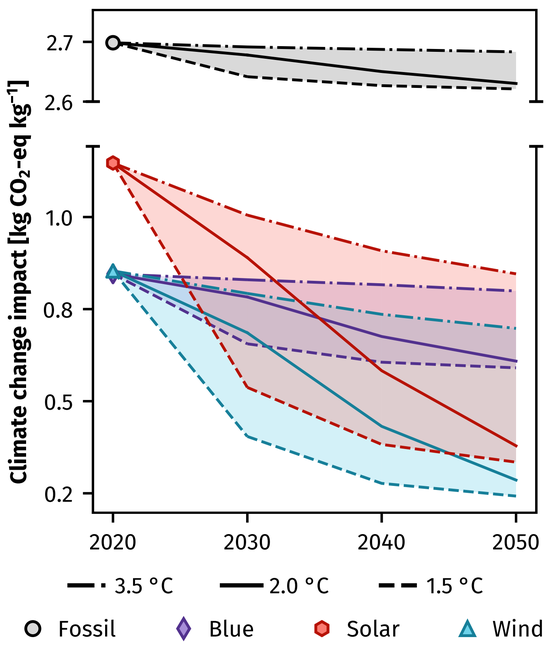


Prospective life cycle assessment!

Approach



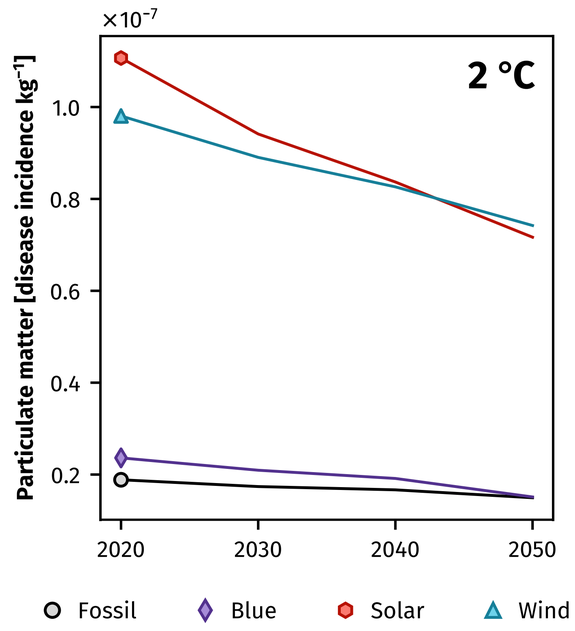
1 Climate change impacts



- The green routes (solar and wind) could produce ammonia with 90% fewer emissions in the 1.5 °C scenario compared to the emissions from the fossil scenario in 2020.
- The reduction is primarily driven by decreased consumption and further decarbonisation of renewable electricity.
- Fossil-based scenarios show a reduction of less than 10%, even in the best case, *i.e.*, the 1.5 °C scenario.
- Blue ammonia, while currently having a smaller impact than green routes, is projected to be outperformed by green ammonia in future scenarios.

90% reduction in emissions can be achieved by 2050

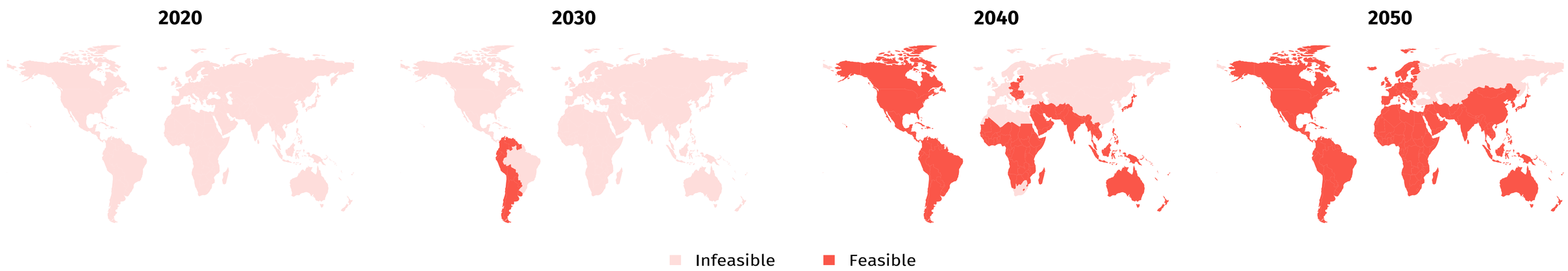
2 Burden shifting



- Analysis extended beyond carbon footprint to assess the potential risk of burden shifting, *i.e.*, worsening other impacts while attempting to combat climate change.
- Green routes for ammonia exhibit higher particulate matter impacts compared to fossil routes, indicating burden shifting.
- Renewable electricity, particularly from solar panels and wind turbines, is identified as a primary source of particulate matter impacts.
- The severity of the burden shifting tends to become less pronounced over time for the green routes.

Burden shifting exists, albeit less pronounced

3 Regional roadmaps (solar ammonia, 2 °C scenario)



- We define feasibility as achieving an 80% reduction in climate change impact compared to the fossil-based scenario in 2020.
- In 2020, none of the regions worldwide met this criterion. By 2050, all regions worldwide are expected to meet the set goal, except for Russia, Turkey, and Central Asia.
- The primary reason for the variability among regions is the diverse power mixes deployed worldwide.

Regions worldwide evolve differently

Conclusions and Outlook

- Transitioning to green pathways for chemical production, combined with ambitious climate policies, can significantly reduce environmental impacts.
- These distilled insights provide tools for researchers, policymakers, industry leaders, and investors to encourage the green production of chemicals.

References [1] *Curr. Opin. Chem. Eng.* **2023**, 39, 100881

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